Automated Eligibility Criteria Extraction

Current Technical Architecture, Semantic Outputs, and Downstream Integration

Modern AutoCrit Project

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1 Purpose and Current Scope

Modern AutoCrit is a standalone local browser application that extracts clinical-trial eligibility criteria from a ClinicalTrials.gov record, pasted eligibility text, or an uploaded full protocol PDF. It produces human-reviewed, validated structured criteria for protocol review, tabular analysis, cohort-definition work, and other downstream clinical data-science systems.

The current output is deliberately richer than a flat Entity–Attribute–Value record. Each atomic criterion separates its clinical domain and entity from its comparator, value, unit, interval, negation, temporal restriction, repetition, qualifier, Boolean relationship, and verbatim source. This separation prevents phrases such as "blood pressure $> 145/95$ on three occasions" from collapsing into one opaque entity string.

The current application does *not* include the retired desktop interface, the deleted clinical-distance screen, the former candidate-attribute critic/reconciliation system, or a full Chia annotation graph. Model output remains a review artifact rather than clinical ground truth. Interrupted model batches are recoverable, but recovery does not replace human review.

2 Lineage and Design Changes

The project began from the protocol-to-structure idea demonstrated by AutoCriteria [1, 2]. The implementation now differs in several ways:

- LangChain orchestration was replaced by a small provider interface and the OpenAI SDK.
- The original long script was divided into registry, LLM, workflow, normalization, persistence, HTTP, and browser modules.
- Desktop interfaces were replaced by a local, full-width web application.
- ClinicalTrials.gov API v2 and full-protocol PDF ingestion were added.
- Human decisions are stored in timestamped JSON and CSV snapshots instead of silently changing the immutable packaged base. The current library can also be downloaded as a formatted Excel workbook for labeling and analysis.
- The former EAV output was replaced by a Chia-inspired structured-criterion projection. Chia distinguishes domain entities, value fields, semantic constructs, and relations; Modern AutoCrit adopts the useful distinctions in a compact tabular form without claiming full annotation-graph compatibility. See https://www.nature.com/articles/s41597-020-00620-0.
- Explicit interval endpoints were added for exact downstream comparison. Equivalent range expressions can therefore share one computational representation.

3 User Workflow

The browser contains three panels.

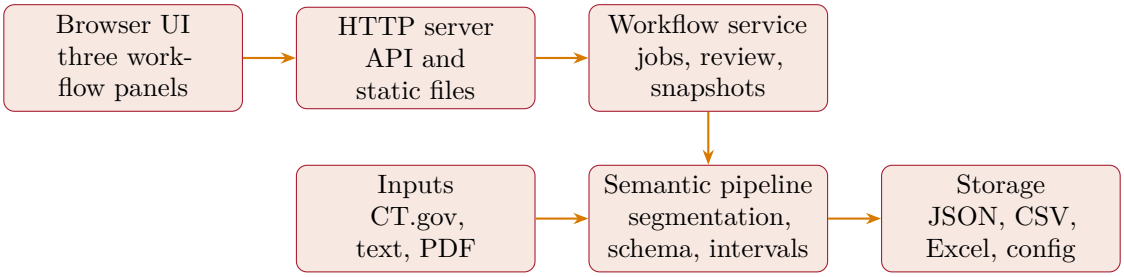
Extraction / ingestion. Select an NCT identifier, paste eligibility text, or upload a complete protocol PDF. PDF ingestion searches for the actual inclusion and exclusion sections and excludes later protocol material such as the SAP and schedule of activities.

Human review. Sort the extracted rows, accept correct rows, or reject a row to edit its domain, entity, comparison, interval endpoints, semantic constructs, and Boolean relation. A corrected row must be saved before the collection can be applied.

Criteria library. Search the immutable examples and the latest reviewed snapshot, download the combined current library as Excel, or export only validated human-reviewed records under the frozen downstream JSON contract. A confirmed reset removes generated review snapshots and returns the display to the packaged examples. Historical EAV snapshots are upgraded in memory for display; their original files are not rewritten unless the user explicitly resets the library.

During extraction, the progress panel reports protocol loading, the number of located source criteria, completed extraction batches, retries, semantic normalization, completion, or failure.

4 Architecture



Component	Responsibility
`main.py`	Launches the browser application or runs a recoverable NCT manifest.
`webapp/server.py`	Routes the JSON API, limits request bodies, serves static assets safely, and starts the threaded local server.
`webapp/static/`	Implements ingestion, progress polling, sortable review, inline correction, and library search.
`backend/services/ctgov_v2_service.py`	Validates NCT identifiers, retrieves API v2 study JSON, and returns eligibility text.
`backend/services/llm/`	Provides the provider-neutral boundary, OpenAI Responses API calls, strict JSON parsing, retry behavior, and token accounting.
`backend/eav_pipeline.py`	Retains its historical filename but now owns the structured-criterion schema, source segmentation, terminology normalization, domain inference, interval parsing, semantic constructs, and relation labels.
`backend/criterion_validation.py`	Checks intervals, negation, units, temporal and repetition constructs, and compound scope; derives validated-export readiness and blockers.
`backend/batch_manifest.py`	Runs or resumes a cohort of NCT IDs while retaining manifest and per-batch checkpoint state.
`scripts/run_trial_cohort.py`	Validates, starts, or resumes the packaged 100-slot cohort from a terminal, with a dry-run check and an optional unattended mode.
`config/trial_cohort_100.json`	Defines a stable cohort identity and 100 active public trials: 50 oncology and 50 cardiovascular records.

Component	Responsibility
`config/trial_cohort_calibration_10.json`	Defines a balanced five-oncology/five-cardiovascular developer subset that exercises the production pipeline without starting the web application.
`backend/services/eav_workflow.py`	Persists jobs and checkpoints, extracts PDF sections, batches model calls, records provenance and progress, upgrades historical snapshots, validates review decisions, and writes criteria-library and validated downstream exports.
`backend/schemas/validated_criteria_v1.schema.json`	Freezes the downstream criterion handoff contract.
`config/terminology_normalization.json`	Governs clinical aliases, exact-only ambiguous abbreviations, and mappings from legacy entity classes to Chia-style domains.
`config/base_criterion_library.json`	Supplies immutable structured seed examples.
`runtime/criteria/`	Stores ignored local job records and runtime library copies.

5 Extraction and Normalization Sequence

1. The browser posts the selected input mode and its content to `/api/extractions`.
2. The workflow validates required input. NCT mode retrieves ClinicalTrials.gov; PDF mode extracts text and locates the actual eligibility section; manual mode uses the pasted text.
3. `segment_criteria()` identifies inclusion and exclusion headings, joins wrapped lines, separates numbered or bulleted criteria, splits independent semicolon-delimited cutoffs, and assigns content-derived parent-statement identifiers.
4. Criteria are sent in bounded batches through `StructuredJsonAdapter` under the strict structured-criterion schema.
5. The model separates coordinated clinical concepts and proposes domains, values, units, temporal phrases, repetition, qualifiers, negation, and AND/OR relationships.
6. `normalize_rows()` rejects invented source strings, restores authoritative inclusion/exclusion labels, adds missing source rows, applies deterministic literal parsing, normalizes terminology, generates stable atomic identifiers, preserves adjacent context, and records Boolean-group membership.
7. Automated checks attach structured findings and withhold invalid or unreviewed criteria from validated downstream export.
8. The browser presents every row for explicit human acceptance or correction.
9. Applying review writes cumulative, versioned JSON and CSV library snapshots to runtime storage and `config/`.

6 Structured Criterion Model

6.1 Domains, fields, constructs, and relations

The compact model adapts four useful aspects of Chia:

- **Domain:** observation, condition, person, device, drug, visit, procedure, or measurement.
- **Entity:** the canonical clinical concept only, without its cutoff or modifier.
- **Fields and constructs:** comparator, value, unit, interval bounds, negation, temporal restriction, repetition, and qualifier.
- **Relations:** derived labels such as `HAS_VALUE`, `HAS_NEGATION`, `HAS_TEMPORAL`, `HAS_MULTIPLIER`, `HAS_QUALIFIER`, `AND`, and `OR`.

Rows from the same source retain a shared `source_id`. This preserves the relationship between atomic concepts split from one sentence.

6.2 Intervals

A numerical rule contains nullable lower and upper bounds with separate inclusivity flags. The application also emits a readable interval string. Examples include:

Source	Interval	Interpretation
`age>18`	`(18,+inf)`	Open lower endpoint.
`age<=65`	`(-inf,65]`	Closed upper endpoint.
`aged18to65`	`[18,65]`	Closed bounded range in years.
`between18and65years`	`[18,65]`	Equivalent normalized range.
`age>=18and<=65years`	`[18,65]`	Equivalent explicit comparator form.

Ratios such as blood pressure `145/95` remain literal values rather than scalar intervals. This avoids pretending that a compound measurement is one scalar cutoff.

6.3 Illustrative structured output

For "Blood pressure $>$ 145/95 on three occasions," the output separates:
- domain: `measurement`; entity: `BloodPressure`;
- comparator: `>`; value: `145/95`;
- repetition: `onthreeoccasions`;
- relations: `HAS_VALUE` and `HAS_MULTIPLIER`;
- verbatim source and stable source identifier.

7 Terminology Governance

The terminology configuration combines three legacy Modern AutoCrit maps with current additions. All legacy attribute and disease aliases are retained. Legacy entity labels are adapted into the current domains rather than treated as disease synonyms. For example, "lab test" maps to `measurement`, "medication" to `drug`, and "diagnosis" to `condition`.

Oncology coverage includes common diseases, performance measures, laboratory concepts, response concepts, and biomarkers, including NSCLC, SCLC, AML, CLL, DLBCL, ECOG performance status, Karnofsky performance status, RECIST, ANC, ALT, AST, ULN, EGFR, ALK, HER2, PD-1, PD-L1, KRAS, BRAF, BRCA1, and BRCA2.

Ambiguous short forms such as `ALL`, `MM`, `CR`, `PR`, `SD`, and `PD` are exact-match only. Thus an extracted entity equal to `ALL` can normalize to Acute Lymphoblastic Leukemia without rewriting the ordinary phrase "all patients."

8 Function Reference

`run_browser_app()`
Creates the threaded server, optionally opens the browser, and closes server resources on interruption.

`BrowserApplication.dispatch()`
Routes health, extraction submission/status, review application, and library requests.

`fetch_study()/eligibility_text_from_study()`
Retrieve one ClinicalTrials.gov API v2 record and its eligibility block.

`EavWorkflowService.submit()/get()`
Validate input, persist a queued job, start its worker, and retrieve its durable JSON state. The historical class name remains for compatibility.

`EavWorkflowService.resume()/list_jobs()`
Restart an incomplete job from its last completed batch and expose saved jobs for later browser review.

`EavWorkflowService._eligibility_from_pdf()`
Read a complete PDF, find the actual eligibility section, remove page furniture, and stop before later numbered sections.

`EavWorkflowService._run()`
Coordinate segmentation, batched model calls, retries, normalization, progress, completion, and failure capture.

`EavWorkflowService.library()`
Load the immutable base and latest reviewed snapshot, upgrading historical EAV rows in memory.

`EavWorkflowService.library_workbook()`
Generate an on-demand Excel export with filterable criteria rows and a field guide.

`EavWorkflowService.validated_export()`
Write and download only human-reviewed, validation-passing criteria under the frozen downstream schema.

`EavWorkflowService.reset_library()`
Delete only generated criteria-library and historical EAV snapshot files under the governed runtime/config locations, preserving the immutable examples and unrelated files.

`EavWorkflowService.apply_review()`
Validate complete human review and atomically write cumulative JSON/CSV snapshots.

`segment_criteria()`
Create inclusion/exclusion source items and stable source identifiers.

`extraction_prompt_for_items()`
Build the source-bounded AutoCriteria/Chia-inspired extraction instruction.

`deterministic_criterion()`
: Parse literal comparison and range syntax, separate semantic constructs, normalize the entity, infer a domain, and derive relation labels.

`canonical_term()`
: Apply exact and phrase-safe aliases while protecting ambiguous abbreviations.

`infer_domain()`
: Use governed domain aliases and deterministic clinical cues when a valid model domain is unavailable.

`interval_notation()`
: Serialize open, closed, bounded, and unbounded numerical constraints.

`semantic_relations()`
: Derive compact Chia-style relationship labels from the structured fields.

`normalize_rows()`
: Reconcile model rows with authoritative sources and produce final review rows.

`validate_criterion()/readiness()`
: Produce machine-readable semantic findings and determine whether a reviewed row may enter the validated export.

`run_manifest()`
: Process or resume an NCT cohort and persist its job mapping and retry state.

`StructuredJsonAdapter.generate_json()`
: Request schema-constrained model output and parse a JSON object.

`OpenAIProvider.generate()`
: Call the Responses API with retry/backoff and token recording.

`atomic_write_json()`
: Write through a temporary file and atomic replacement.

9 Outputs and Their Meaning

9.1 Job record

Each submission creates `runtime/criteria/jobs/<job_id>.json`. It contains the job, trial, and protocol identifiers; source version, URI, retrieval time, and protocol hash; input mode; timestamps; status; stage; progress; error; source items; completed model batches; checkpoint rows; final rows; and an ordered event log.

Status	Meaning
`queued`	The durable job record exists and its worker has not started.
`running`	Ingestion, model extraction, or semantic normalization is active.
`completed`	Structured rows are available for human review; they are not yet governed additions.

Status	Meaning
`failed`	The job stopped and `error` contains the immediate exception.

9.2 Review rows

Every row contains stable criterion and parent-statement identifiers, protocol/source version, source and surrounding context, Boolean-group membership and atom order, criterion type, domain, entity, comparison fields, semantic constructs, model extraction confidence, validation findings, human-review status, and validated-export readiness. Human rejection opens editable semantic fields; saving distinguishes a corrected row from one accepted unchanged.

9.3 Criteria-library snapshots

Applying review creates timestamped `criteriaLibrary` JSON and CSV snapshots. JSON is authoritative and uses structured-criteria schema v4; CSV provides a convenient tabular export. Copies are written to `runtime/criteria/libraries/` and `config/`; generated config copies are ignored by Git.

The Criteria Library panel also exposes `/api/library.xlsx`. This endpoint generates a workbook from the immutable base plus the latest reviewed snapshot without changing either source. The main worksheet contains normalized fields, source evidence, frozen headers, and filters; a second worksheet explains the fields for human labelers.

The `/api/validated-export.json` endpoint emits only rows with complete provenance, accepted or corrected human review, and no blocking validation findings. Its frozen schema identifier is `modern-autocrit.validated-criteria.v1`; a copy is retained under `runtime/criteria/exports/`.

The reset endpoint requires browser confirmation and removes generated `criteriaLibrary_*` and historical `eavLibrary_*` JSON/CSV snapshots from the runtime library directory and `config/`. It does not remove extraction job records, unrelated files, or `base_criterion_library.json`. The immutable records use trial marker `EXAMPLE` and include deliberately difficult cases for interval boundaries, sex wording, ambiguity, temporal and repeated measurements, exceptions, and compound logic.

The application can read historical `eavLibrary_*.json` snapshots. It converts their comparator/value form into the richer representation in memory and does not mutate the old file. The first newly applied review carries the upgraded records into a v4 criteria snapshot. Legacy records without adequate provenance remain visibly ineligible for validated export.

10 Errors and Operational Interpretation

Error or symptom	Meaning and response
Missing API key	Add `OPENAI_API_KEY` to the repository-root `.env` and restart.
Invalid or missing NCT ID	Check the identifier and registry availability.
No eligibility criteria segmented	Headings or list formatting were not recognized; use manual text or inspect the PDF section.

Error or symptom	Meaning and response
PDF section not recognized	The document lacks recognizable inclusion/exclusion headings or is image-only. OCR is not currently performed.
Structured response failure	The provider did not return the strict schema after retries. Completed batches remain checkpointed; resume the job or rerun its manifest.
Request body too large	The server accepts at most 75 MB JSON bodies and the browser limits PDF selection to 55 MB.
Review cannot be applied	Every row must be accepted or corrected and pass the automated interval, negation, unit, temporal, repetition, and compound-scope checks.
Port already in use	Stop the prior process or launch with a different `--port`.
Legacy snapshot shown	No v4 snapshot exists yet; the newest historical file is being upgraded in memory. Apply a reviewed run to create v4 files.

11 Current Limitations

- Jobs and completed batches are persisted and resumable, but prompts and raw model responses are not yet written as a complete audit trail.
- PDF extraction is text-based and does not OCR scanned protocols.
- The criteria library stores reviewed trial assertions. A future governed concept layer should separate reusable concept definitions from trial-specific constraints.
- Terminology normalization is a curated alias system, not an ontology service. RxNorm, LOINC, OMOP, NCIt, or ICD links are not yet assigned.
- The representation is inspired by Chia but does not implement arbitrary nested annotation graphs or executable patient queries.
- Cross-trial distances, calibration, and learned representations are outside this extractor. The validated schema can supply their inputs.

12 Packaged 100-Trial Cohort Workflow

The repository includes a terminal workflow for producing a research dataset without using the browser to start each extraction. The manifest `config/trial_cohort_100.json` has exactly 100 numbered, active slots. Slots 1–50 contain oncology NCT identifiers selected from public ClinicalTrials.gov interventional records with reported eligibility text. They span solid tumors and hematologic malignancies, multiple intervention classes, and several trial phases. Slots 51–100 contain cardiovascular trials spanning heart failure and cardiomyopathy, atrial fibrillation, coronary and valvular disease, hypertension and pulmonary hypertension, stroke, thrombosis, peripheral arterial disease, and aortic aneurysm. This deliberate breadth supports within-area and cross-area comparisons without limiting the corpus to one narrow disease pattern.

Run the following commands from the repository root:

```
./scripts/run_trial_cohort.py --calibration --dry-run
./scripts/run_trial_cohort.py --calibration
```

```
./scripts/run_trial_cohort.py --dry-run
./scripts/run_trial_cohort.py
```

The calibration option runs five oncology and five cardiovascular trials through the same production extraction path without launching the browser. The equivalent lower-level developer entry point is `pythonmain.py--calibration`. Its manifest intentionally uses the full cohort's stable `cohort_id`; therefore, completed calibration jobs are recognized and skipped when the operator later runs all 100 trials. Human review may use either the browser or the spreadsheet-only path below.

After extraction or validation logic changes, an intentional clean calibration rerun is started with `./scripts/run_trial_cohort.py--calibration--fresh`. This creates new jobs and updates the active manifest records while retaining prior job JSON files for audit and comparison.

When saved model responses remain valid and only deterministic normalization or validation changes, the following command rebuilds completed rows directly from their checkpoint payloads and creates a replacement workbook:

```
./scripts/run_trial_cohort.py --calibration --renormalize
```

This path makes no registry or model API calls.

The dry run checks the selected manifest shape, confirms its active identifiers, detects duplicates, and makes no registry or model calls. The normal script command displays a cost warning and requests confirmation. For a supervised terminal or job runner, `--yes` bypasses only that confirmation; it does not bypass validation, retries, or human review.

The batch service processes active NCT identifiers sequentially. Each trial still uses the same protocol retrieval, source segmentation, bounded model batches, normalization, and validation path as an interactive extraction. Completed model batches are checkpointed. The manifest state maps each NCT identifier to its persistent job and is stored under `runtime/criteria/manifests/`. The stable manifest `cohort_id`, rather than the mutable file checksum, names this state, allowing calibration work to carry into the full run. A content fingerprint is retained in the state for auditability.

A completed batch run is not a completed research dataset. The cohort script automatically writes completed jobs to a timestamped human-review XLSX in the repository root. Its first sheet, *Validation Instructions*, contains review guidance plus cohort-wide, per-trial, and per-finding quality metrics. The second sheet, *Human Review*, begins with verbatim source evidence and the complete parent statement, followed by editable semantic fields, automated findings, confidence, and then provenance and identifiers. Freezing at `C2` keeps both evidence columns visible, and protocol URLs are clickable. Reviewers mark every row `accepted` or `corrected`; corrected rows are edited directly in the sheet. The command `--export-review` regenerates this workbook without registry or model calls, while `--apply-review<xlsx>` validates and applies only trials whose rows are all reviewed. Thus, human review may be completed entirely through the terminal and spreadsheet, although the browser remains available. After applying review, the operator resolves any reported blockers and freezes the validated JSON export with the cohort manifest, schema version, and source-control revision. Failed trials remain documented in manifest state and do not prevent later trials from running.

12.1 Two Complete Script-Only Runs

The 10-trial calibration workflow is:

```
./scripts/run_trial_cohort.py --calibration --dry-run
./scripts/run_trial_cohort.py --calibration
./scripts/run_trial_cohort.py --calibration --export-review
# Edit and save the generated XLSX.
./scripts/run_trial_cohort.py --calibration \
  --apply-review <calibration-review.xlsx>
```

The full 100-trial workflow is:

```
./scripts/run_trial_cohort.py --dry-run
./scripts/run_trial_cohort.py
./scripts/run_trial_cohort.py --export-review
# Edit and save the generated XLSX.
./scripts/run_trial_cohort.py \
  --apply-review <full-cohort-review.xlsx>
```

Neither workflow starts the HTTP server or browser. Both are recoverable: rerunning extraction resumes or skips saved jobs, and review application skips incomplete trials. The extraction command automatically creates a workbook at normal completion; the explicit export command is retained so a reviewer can capture all currently completed jobs after an interruption or resumed run.

13 Optional MetricSpace Integration

The structured output permits distinct comparison channels:

- canonical-entity agreement and governed synonymy;
- domain compatibility;
- exact interval relations, including equivalence, subset, superset, overlap, and disjointness;
- comparator direction and endpoint sensitivity;
- negation, inclusion/exclusion polarity, temporal scope, repetition, qualifier, and Boolean-structure disagreement;
- embedding or LLM judgment for residual clinical semantics that cannot be resolved structurally.

Reviewed human corrections can become labeled supervision. A future agentic layer can select structural comparison for exact constraints, retrieve terminology candidates, invoke an embedding model for paraphrases, request LLM adjudication for ambiguity, reconcile disagreement, quantify uncertainty, and escalate unresolved cases.

Modern AutoCrit is intentionally near feature freeze. For a MetricSpace study, remaining work is operational data production: select trial cohorts, run recoverable extraction, complete human review, resolve validation blockers, and freeze versioned validated exports. Raw model audit records, ontology identifiers, and a separate governed concept layer remain possible later improvements rather than prerequisites for that downstream project.

14 Conclusion

Modern AutoCrit is now a standalone browser-delivered protocol eligibility extractor and human-review tool whose output is designed for analysis and reuse rather than display alone. Its v4 library schema and frozen validated-criteria v1 handoff preserve identifiers, source versions, context, quantitative intervals, semantic constructs, Boolean scope, review state, automated validation, and readiness. Specialized comparison or representation-learning systems can consume this output without being required to understand Modern AutoCrit internals.

References

[1] S. Datta, *AutoCriteria*, GitHub repository.
https://github.com/surdatta/autocriteria

[2] S. Datta et al., *AutoCriteria: A Generalizable Clinical Trial Eligibility Criteria Extraction System Powered by Large Language Models*, 2023.
https://arxiv.org/abs/2304.06096